# Sushant Mudalgi

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## SUMMARY

Software engineer with four years of experience building distributed data systems, backend services, full stack applications, and AI systems on AWS, Azure, and GCP, including an agentic RAG system with evals and guardrails.

## EDUCATION

**Northeastern University, Khoury College** - MS in Computer Science — Aug 2025 - May 2027

*Coursework:* Algorithms, Programming Design Paradigm, Foundations of AI, Advanced Database Management Systems

**KLE Technological University, Hubli** - B.E. Computer Science; **GPA: 3.91/4.00** — Aug 2017 - May 2021

## SKILLS

**Languages**: Python, Java, TypeScript, JavaScript, SQL, C++, C#, Bash

**Backend and Frontend**: REST APIs, Microservices, FastAPI, Django, Node.js, React.js, Angular, PostgreSQL, MongoDB

**Data and Distributed Systems**: PySpark, Databricks, Snowflake, Amazon Redshift, dbt, Airflow, ETL, Data Modeling

**Cloud and DevOps**: AWS, Azure, GCP, Docker, Kubernetes, Terraform, AWS CDK, CI/CD, GitHub Actions, Jenkins

**AI and ML**: LLMs, RAG, AI Agents, MCP, LangChain, PyTorch, TensorFlow, scikit-learn, Evals

## EXPERIENCE

**Takeda** — May 2025 - Aug 2025

*Software Engineer* — *Bangalore, India*

- Built and operated 15+ production data services on Databricks and AWS (S3, DMS, Step Functions, Lambda) powering regulatory and business reporting across North America and Europe.
- Containerized services with Docker and deployed them on Kubernetes through CI/CD.
- Built React dashboards consolidating Facebook and Google campaign data into self service reporting.
- Implemented access controls, metadata, and lineage with Databricks Unity Catalog for sensitive clinical data.

**EY GDS** — Nov 2024 - Apr 2025

*Software Engineer* — *Bangalore, India*

- Built 15 real time and batch pipelines in Python and PySpark processing 200 to 300 GB daily from five heterogeneous sources, including SQL Server, SFTP, SharePoint, Heroku, and vendor feeds.
- Cut workflow costs 25 percent across 10+ processes by migrating orchestration to Tidal and Airflow.
- Developed LLM automation with the OpenAI API and LangChain that replaced a manual metadata tagging workflow.
- Created full stack internal tools with Django, React, and PostgreSQL that replaced manual log inspection.
- Modeled 30+ Delta Lake tables with dbt across a medallion architecture, with validation checks catching bad records early.

**Deloitte USI** — Sep 2021 - Nov 2024

*Software Engineer* — *Bangalore, India*

- Developed REST APIs and microservices in Java, FastAPI, and Node.js, with React and Angular front ends.
- Migrated legacy ETL from Talend to AWS Glue and Java based frameworks, improving maintainability.
- Implemented pgvector semantic search, and applied PyTorch and TensorFlow models to oncology imaging data.
- Handled security remediation, including RBAC changes and vulnerability fixes, and supported production systems on call.

**Kalpas Innovations** — Mar 2021 - Jul 2021

*Software Engineering Intern* — *Hubli, India*

- Developed the front end of a consumer web platform used by 5,000+ people monthly with React.js, backed by MongoDB.

## PROJECTS

**OpsPilot: Agentic RAG Copilot** [Git] | *Python, FastAPI, Claude, pgvector, AWS CDK*

- Built an agentic RAG system that diagnoses production alerts through a retrieve, investigate, synthesize, and verify loop, returning cited answers with confidence scores and human approval for risky actions.
- Achieved 1.000 hit@5 and 1.000 MRR with hybrid BM25 and dense retrieval fused by Reciprocal Rank Fusion, versus 0.909 MRR for either alone, enforced by a CI evaluation gate.
- Deployed on AWS via CDK with ECS Fargate and RDS Postgres, exposing tools through an MCP server.

**DealRadar: Price Monitoring Dashboard** | *Python, Pandas, PostgreSQL, Streamlit*

- Built a pipeline and dashboard tracking Amazon and Walmart prices over time in PostgreSQL.